Eigen-Decomposition of Artificial Neural Networks

Yogev Angelovici, Juan Carlos Gonzalez, Ethan Martinez, Max Watzky, and Michael Yao

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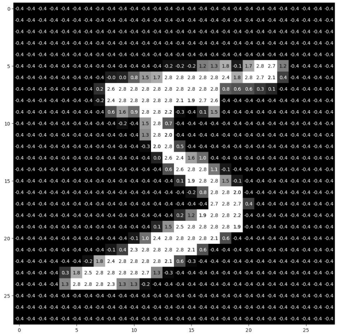
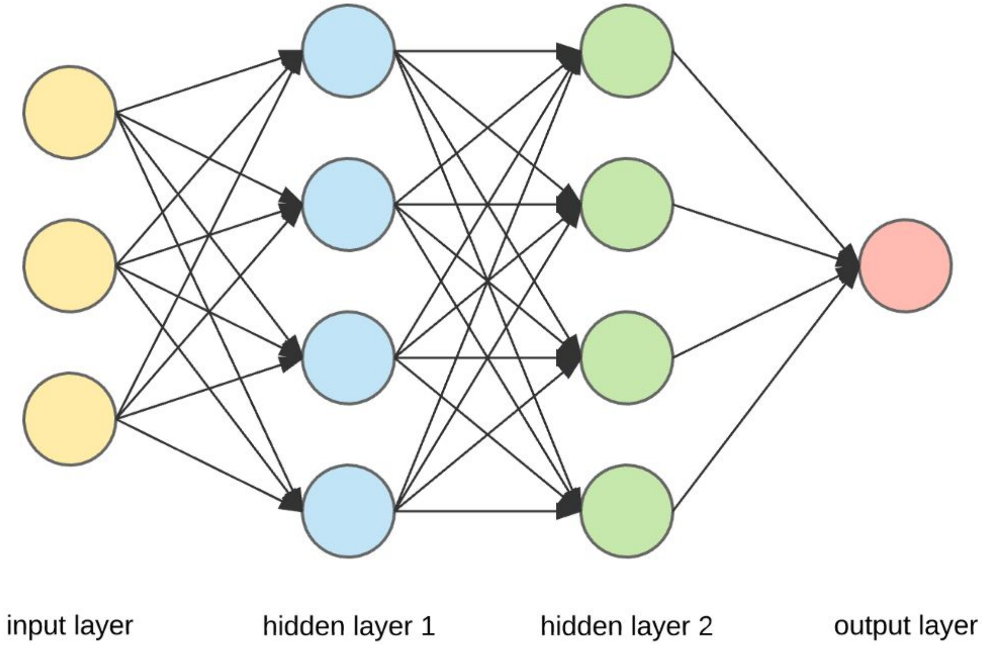
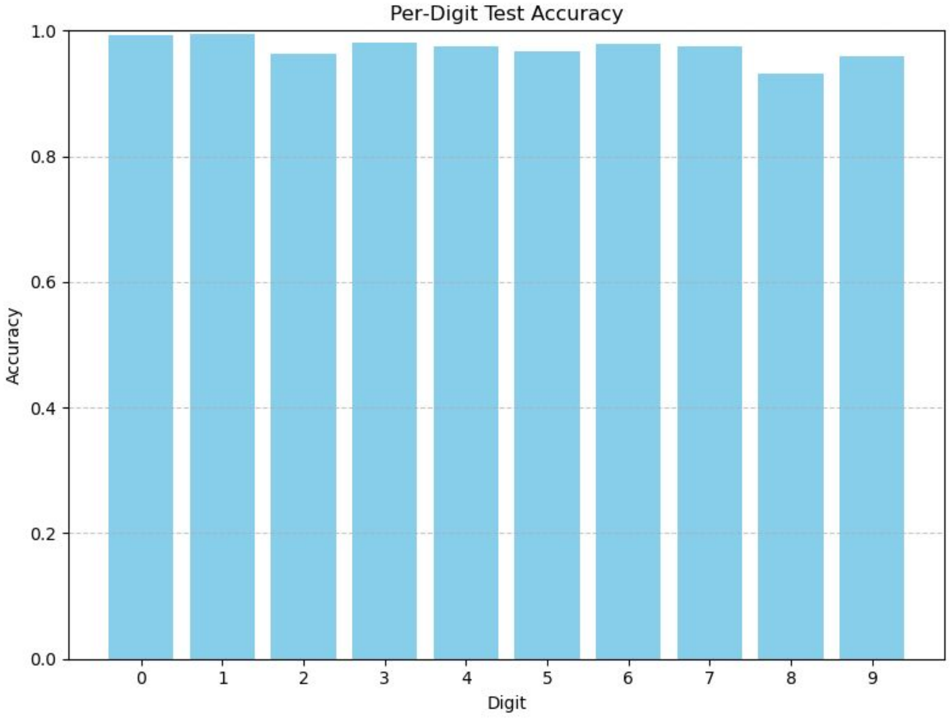
Neural networks are excellent prediction devices

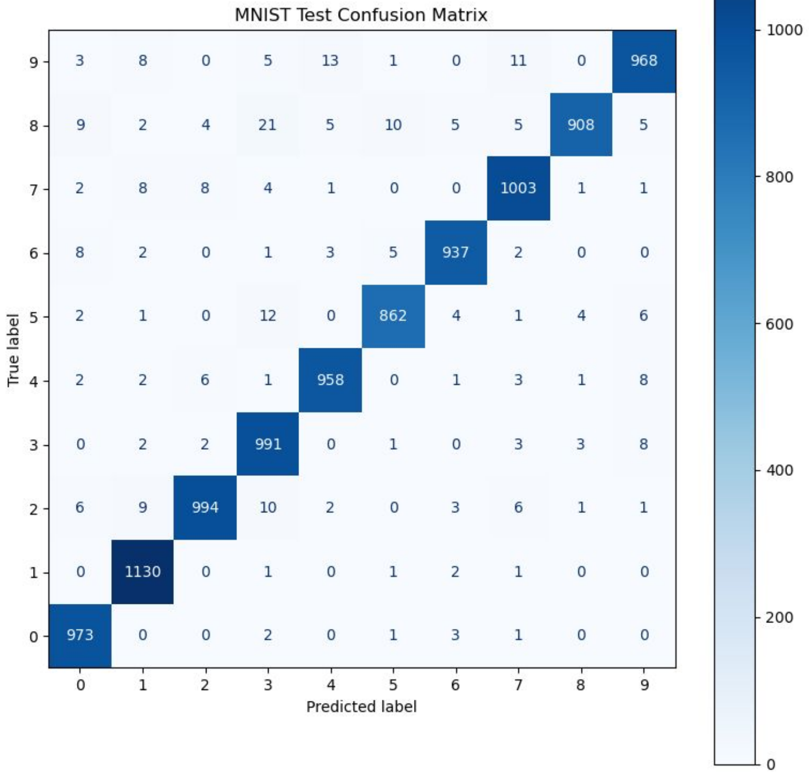




99.8% accuracy: **Digit 5**

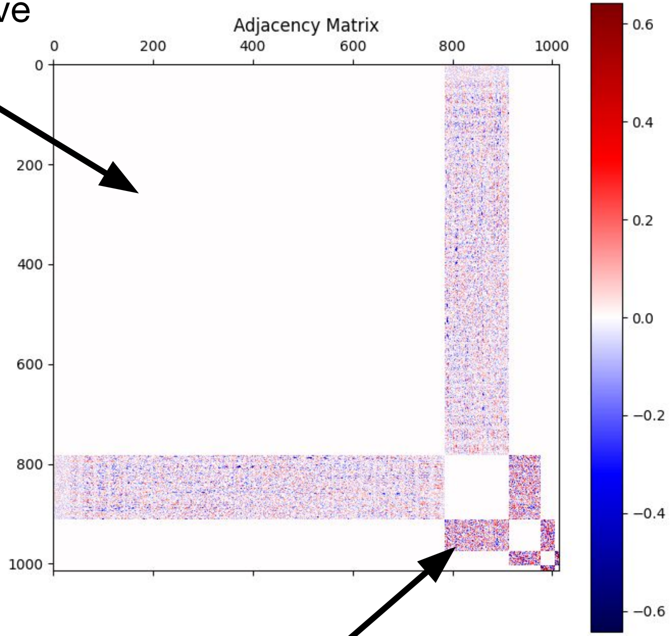
Neural networks are excellent prediction devices





Neural network parameters are not readily interpretable

Unconnected neurons have weight “0”



Connected neurons have finite weight

Network Adjacency Matrix

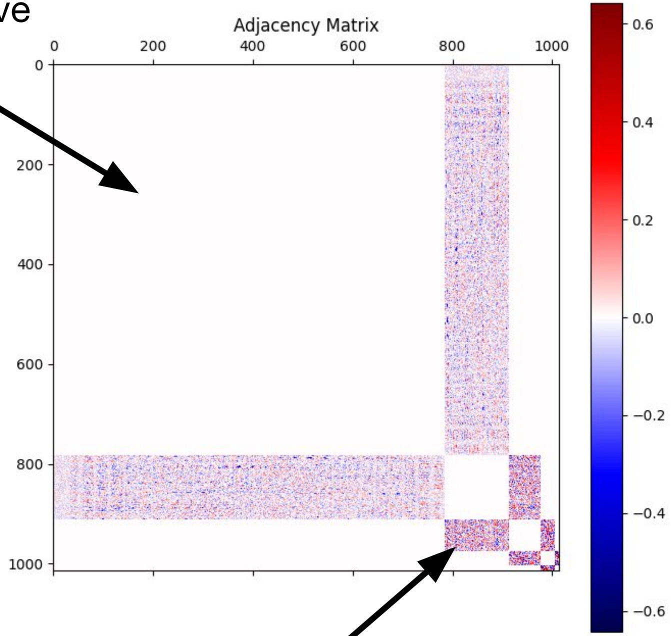
- Shows how different neurons feed into one another
- Blocks represent layers

But this looks meaningless!

- How can we interpret what the NN thinks is “one-ness” or “five-ness”?

Neural network parameters are not readily interpretable

Unconnected neurons have weight “0”



Connected neurons have finite weight

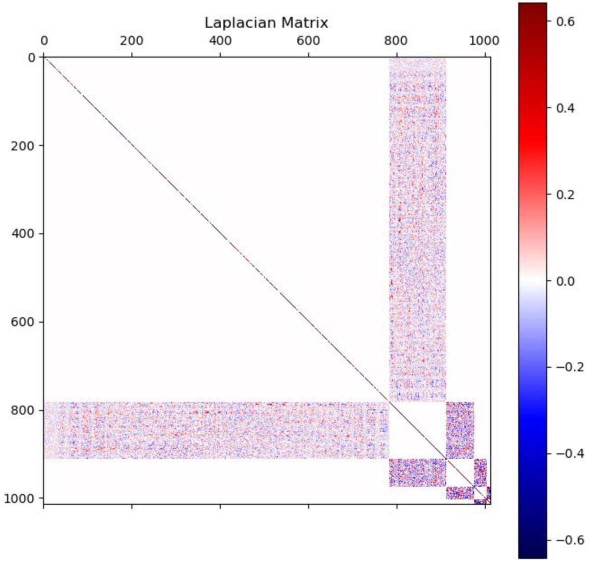
Idea: Individual weights look meaningless, but together, they must form **ensembles** that encode higher dimensional patterns

- Digits with “holes” (like 8, 6, and 9)
- Digits with “straight lines” (like 1 and 7)
- Etc.

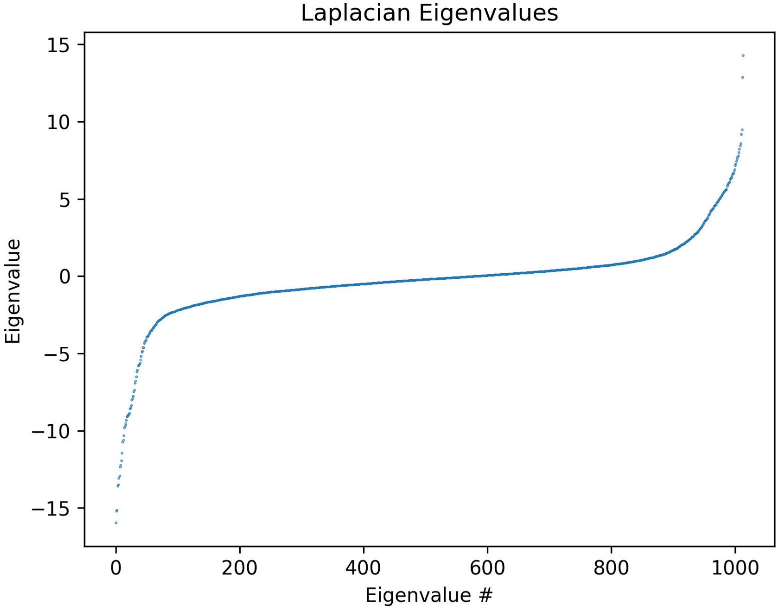
1. Can we find a NN decomposition that reveals the “intuition” behind its predictions?

2. Can we use this decomposition to reconstruct a more efficient NN?

Eigendecomposition of the NN Laplacian



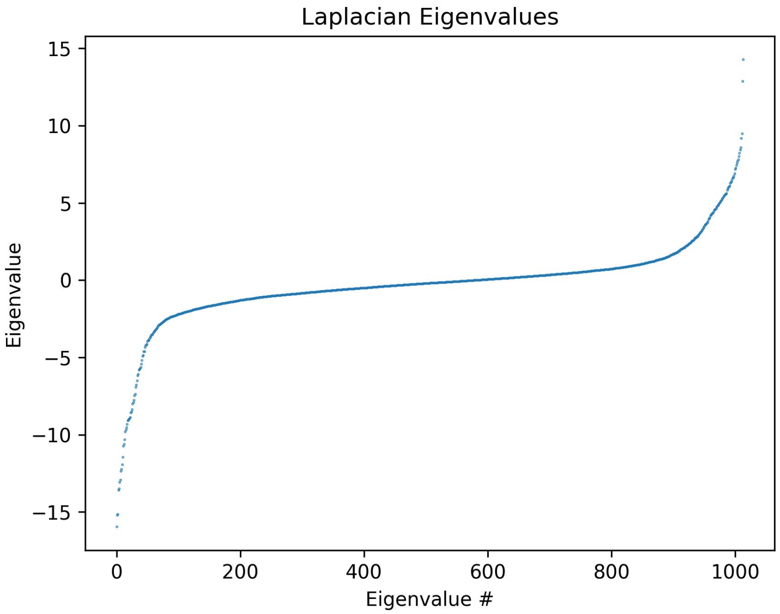
$$\mathbf{A} = \mathbf{PDP}^{-1}$$



Eigendecomposition of the NN Laplacian

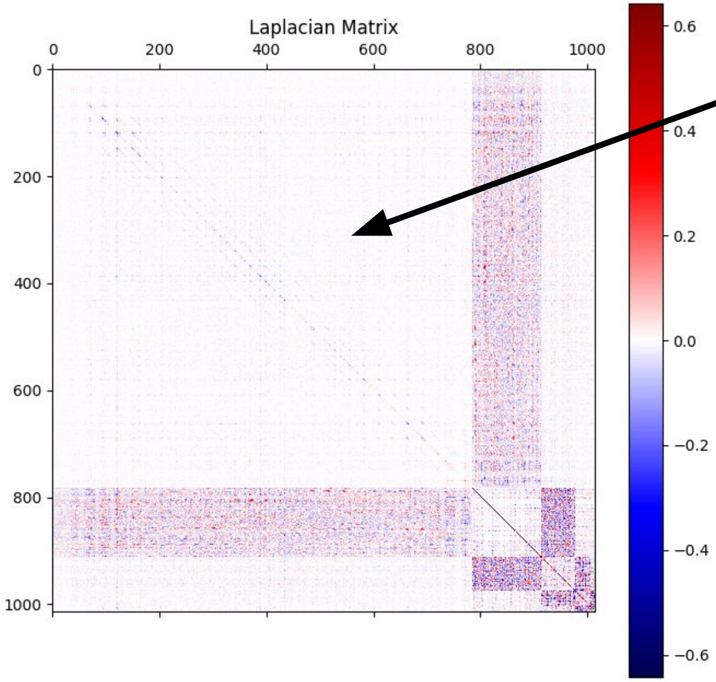
Most eigenvalues seem to do **very little,** but a few are **extremely influential**

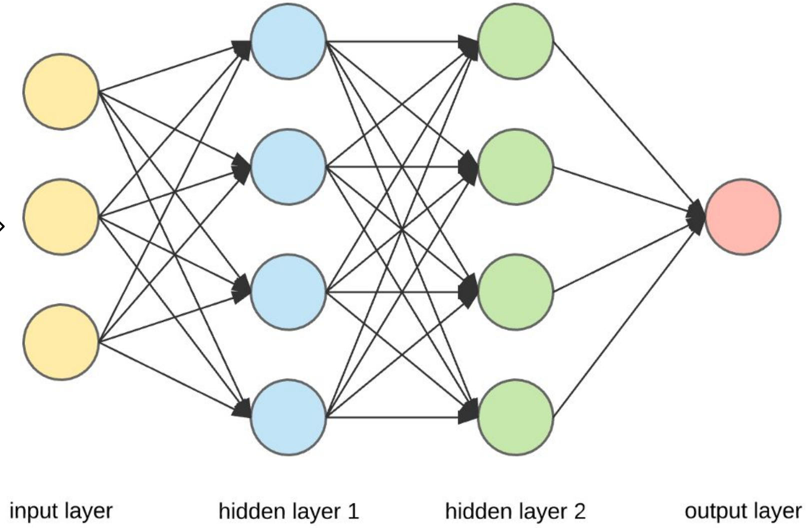
Idea: Sort the eigenvalues by absolute value, set the lowest ranked eigenvalues to 0, and reconstruct the neural network



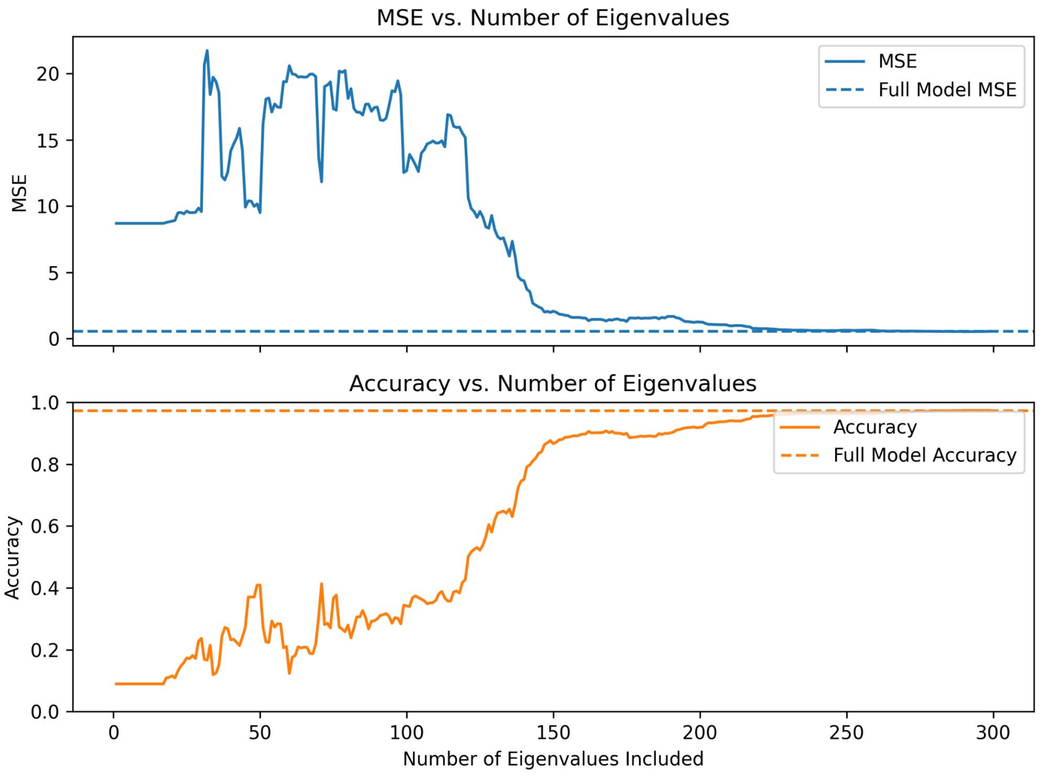
Low-rank eigen-reconstruction

Artifacts in previously empty regions are ignored





Reconstruction enables massive compression



Only **~140/1014** eigenvalues are necessary to achieve reasonable accuracy/loss!

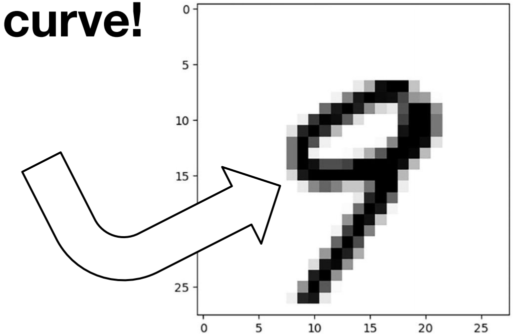
Opens possibilities for **model compression** to a small fraction of the original size

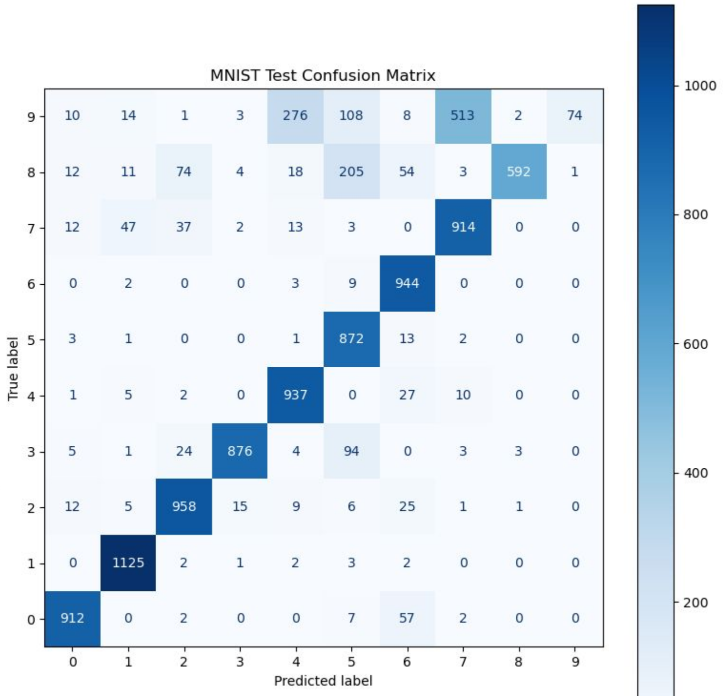
But what's actually going on when we remove eigenvalues?

Groups of eigenvalues encode meaningful features

Removing eigenvalues ranked 100-150 **ONLY impacts recognition of the digit "9"**

- 9 is now mistaken for 7
- These eigenvalues might be responsible for detecting a **single curve!**



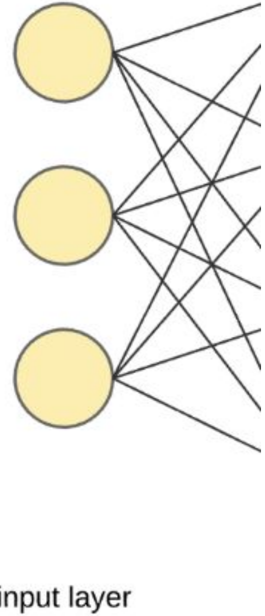


Question: Can we do this with individual layers?

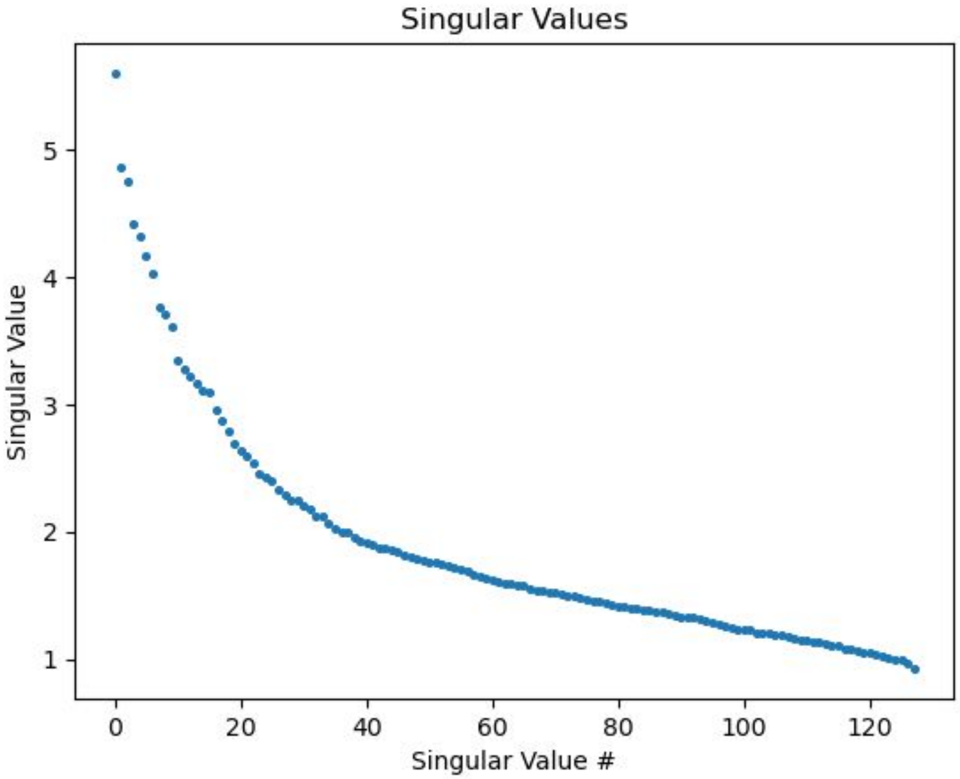
Problem: These layers are **not** square matrices

Solution: Use SVD!

SVD on the First Layer of Weights

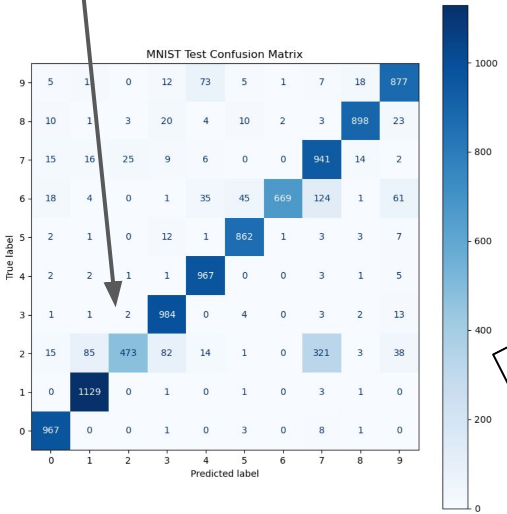


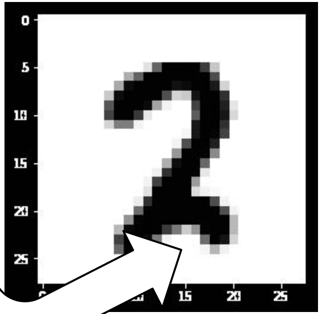
$$\mathbf{M} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^*$$

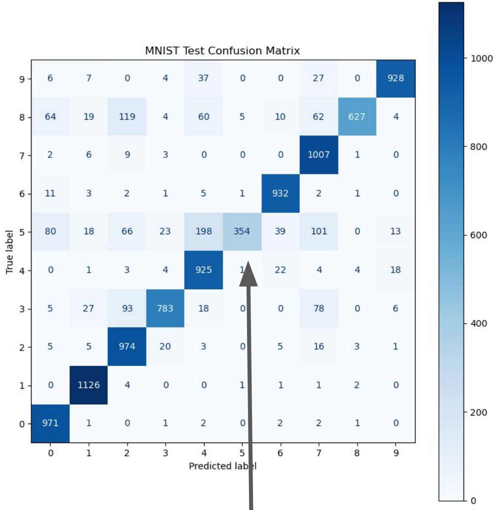


Individual singular values encode meaningful features

Removing the first singular value **makes the model confuse 2 and 7!**







Removing the second singular value **only impacts recognition of the digit 5!**

Takeaways and next steps

Interpretability

Singular directions map to digit parts

Compression

Hundreds of eigenvectors → massive reduction of parameters

Future steps

Try on CNNs/language models

Real-time spectral pruning

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